Research Article

Alternatives to Laboratory Animals
2025, Vol. 53(4) 188–202

DOI: 10.1177/02611929251351559
journals.sagepub.com/home/atl
Sage

Xeno-free alternatives to the use of fetal bovine serum in head and neck cancer explant culture

Alexya Azhakesan[1,*], Elena Seiz[1,*], Johann Kern[1], Laura Hendricks[1], Jens Fleckenstein[2], Frederic Jungbauer[1], Sonja Ludwig[1], Christoph Brochhausen[3], Luis Bugia[1], Claudia Scherl[1], Anne Lammert[1], Nicole Rotter[1] and Annette Affolter[1]

Abstract

Patient-derived head and neck squamous cell carcinoma (HNSCC) explant models have been shown to retain the original tumour microenvironment and morphological characteristics. To enhance the human relevance of models and improve clinical outcomes, there is an emerging move toward preclinical models that are cultured in xeno-free media (i.e. media containing no non-human animal-derived components). Fetal bovine serum (FBS) has been the standard cell culture medium supplement in most *in vitro* systems. However, growing emphasis on ethical concerns, animal welfare considerations and reproducibility, as well as the need to implement the Three Rs principles, have driven substantial efforts to identify viable xeno-free alternatives to FBS. In this study, an *ex vivo* culture model for HNSCC was developed, based on the use of xeno-free media. Human platelet lysate (hPL)-supplemented medium and a commercially available xeno-free human mesenchymal stromal cell (MSC) expansion medium were evaluated, comparing HNSCC explant model growth to that in the 'standard' FBS-supplemented medium, over a 10-day culture period. To best reflect clinical conditions, the tissues were treated with radiochemotherapy (RCT) comprising cisplatin and fractionated irradiation. After 10 days, the tissues were formalin-fixed, paraffin-embedded, and assessed for the expression of key biomarkers, including PD-L1, Ki-67 and vimentin. The up-regulation of PD-L1, as well as the downregulation of Ki-67 and vimentin, were consistent across all media, thus validating hPL-supplemented medium and MSC medium as viable alternatives to FBS-supplemented medium, for use in the culture of humanised HNSCC preclinical models.

Keywords

3Rs, animal free tumour model, *ex vivo*, head and neck cancer, histomorphology, radiochemotherapy, tissue slices

Introduction

Head and neck squamous cell carcinoma (HNSCC) is one of the most invasive and metastatic solid tumours among the head and neck cancers, with limited therapeutic options other than surgical resection.[1,2] HNSCCs are characterised by high intertumoral and intratumoral heterogeneity which is reflected in varying treatment responses, frequent development of resistance, poor prognosis and a subsequent low survival rate.[3,4] Currently, in addition to surgical resection, chemotherapy and radiotherapy, new treatment strategies such as immune checkpoint inhibition (e.g. anti-PD-1 and PD-L1 agents) are increasingly used in clinical practice.[5,6] Despite multimodal therapies, the clinical success rate falls short of expectations. Thus, human-relevant models that recapitulate the native tumour microenvironment are needed, in order to identify robust, predictive and prognostic biomarkers that stratify patient response.[1,5,6]

[1]Department of Otorhinolaryngology, Head and Neck Surgery, University Hospital of Mannheim, Medical Faculty Mannheim, University of Heidelberg, Mannheim, Germany
[2]Department of Radiation Oncology, University Hospital Mannheim, Medical Faculty Mannheim, University of Heidelberg, Mannheim, Germany
[3]Institute of Pathology, University Hospital Mannheim, University of Heidelberg, Mannheim, Germany

*These authors contributed equally to this work and share first authorship.

This article is part of the Virtual Special Collection on *Serum-free Approaches*.

Corresponding author:
Annette Affolter, Department of Otorhinolaryngology, Head and Neck Surgery, Medical Faculty Mannheim, University of Heidelberg, Theodor-Kutzer-Ufer 1-3, Mannheim D-68167, Germany.
Email: Annette.affolter@umm.de

To date, animal models have been extensively used in preclinical phase trials. However, ample research has shown that animal models cannot fully recapitulate human tumours.[1,2,5] This is reflected in the fact that these models can fail to detect the poor efficacy and toxic effects of the drugs being tested, due to their low predictability, thereby contributing to a significant translational gap.[5] Inter-species and intra-species differences that result in variations in drug response, the lack of genetic diversity within inbred laboratory animals, and the overall growing ethical concern over animal testing, have led scientists and policymakers to seek more translatable humanised disease models.[7] Meanwhile, at an *in vitro* level, simple and well-established (but highly non-representative) 2-D models have traditionally been used for expanding our basic understanding of HNSCC pathophysiology, and for studying anti-cancer drug mechanisms, kinetics and toxicity.[8] In the past decade, to circumvent the use of animal models and to shift the research focus toward a native-like complex model system, 3-D *in vitro* modelling has been used (e.g. spheroids, organoids, scaffolds, bioprinted and organ-on-a-chip models).[1,9,10] Such approaches are deemed more physiologically relevant, partly due to their ability to incorporate complex and heterogeneous tumour microenvironment entities.[1,9,10]

Our previous research has involved the establishment of patient-derived (i.e. *ex vivo*) HNSCC tumour explant models. Such models incorporate the complexities of the tumour microenvironment by preserving the native tissue architecture and cellular diversity, while retaining tumour–stroma interactions.[11,12] Also, HNSCC tumour explant models display greater *in vivo* relevance, since they are directly derived from the patient tumour tissue, and can be used in personalised theranostic applications. Previous studies showed that our HNSCC explants remained viable for over 21 days in culture, displaying the characteristic features of invasive, silent and metastatic HNSCC tumours with preservation of morphological features comparable to the 2-D HNSCC model.[13] In line with this work and incorporating our prior knowledge, we attempted to create a 3-D HNSCC tumour explant model based on the use of xeno-free media (i.e. media containing no non-human animal-derived components). It was proposed that this new model with xeno-free media would more closely recapitulate the native situation and thus improve translatability.

Recent advancements in the field have focused on the establishment of humanised models, including fully animal component-free or xeno-free options for culture, assays and biomarker identification. It is well-known that optimised growth media are fundamental to the success of *in vitro* cell and tissue culture models in basic life sciences and translational research. Thus, the use of appropriate growth medium is essential for the study of biological processes, biochemical and mechanical cues, and novel drug activity, toxicity and underlying mechanisms.[14] Traditionally, fetal bovine serum (FBS) has been the standard cell culture medium supplement in many well-established disease models.[15] However, in recent decades, concern over the use of FBS has escalated. Ethical issues regarding animal welfare have been raised, since FBS is normally harvested through the cardiac puncture of fetuses.[16–19] In addition, the intrinsic and unavoidable batch-to-batch variation of FBS can lead to poor experimental reproducibility, contamination risks and potential misinterpretation of therapeutic efficacy due to its xenogeneic properties.[16–19] In line with the Three Rs principles (*replacement*, *reduction* and *refinement*), scientific concern over poor experimental reproducibility has driven the trend toward reducing the use of animal products and promoting the use of animal component-free or xeno-free cell culture media in the development of humanised tumour model systems.[16,17]

This study explores the replacement of FBS with xeno-free alternatives, as growth medium supplements in HNSCC explant culture. In the search for FBS substitutes, human platelet lysate (hPL) has lately emerged as a promising candidate.[18,20,21] The potential of hPL as a xeno-free alternative to FBS lies not only in its human-derived origin, but also in its clinical relevance.[18,20,21] As hPL closely mimics human physiology in terms of blood serum components, it offers a more physiologically relevant environment for culturing human tissues; this potentially leads to more translatable results than obtained with 'traditional' FBS media supplementation, as well as being a more ethically acceptable methodology.[18,20,21] The widespread use of hPL in tissue culture could thus be a promising step toward more clinically relevant and ethically sound research.

Our goal was to evaluate whether hPL-supplemented medium, or a commercially available xeno-free human mesenchymal stromal cell (MSC) expansion medium, could effectively replace FBS-supplemented medium in 3-D HNSCC explant culture, potentially offering a more human-focused approach to tumour modelling. HNSCC explants were cultured for 10 days in the two xeno-free media and then compared to explants grown in FBS-supplemented medium. If it were proven that hPL supplementation consistently supports tissue viability, proliferation and functional outcomes at levels comparable to, or superior to FBS, then it could potentially be promoted as a standard medium supplement for a wide range of human tissue culture applications, including personalised medicine and drug development. However, this would be subject to further clinical and translational evaluation.

The clinical treatment for HNSCC can typically involve immunotherapy targeting PD-L1 and/or radiation administered together with platinum-based chemotherapy, which acts as a radio sensitiser to increase the overall effectivity of the treatment.[3] To best reflect this recommended clinical regimen, the *ex vivo* tissues used were treated with radiochemotherapy (RCT). The RCT-treated HNSCC explants were characterised in terms of their morphology (haematoxylin and eosin (H&E) staining) and the expression of specific markers, including Ki-67 (proliferation marker), PD-L1 (immune checkpoint marker) and vimentin (epithelial to mesenchymal transition (EMT) marker). These markers were selected for their clinical relevance to the evaluation of the therapeutic treatment response. PD-L1 expression reflects immune evasion mechanisms employed by tumour cells and is a key target in checkpoint blockade immunotherapy, which is increasingly used in HNSCC treatment.[3] EMT, indicated by vimentin expression, is associated with tumour progression, contributing to increased tumour invasiveness, therapeutic resistance and metastatic potential in HNSCC.[3] Fresh (i.e. uncultured) primary tissue was also assessed, in order to determine whether there was any evident change in biomarker expression levels as a result of the tissue processing *per se*.

An overview of the step-by-step processes involved in establishing the HNSCC explant model is shown in Figure 1.

Material and methods

Tumour sample collection

Patients with a first diagnosis of a HNSCC tumour were sensitively approached with regard to participating in the study. Informed consent was obtained from the patients who agreed to participate in the study (Ethics Vote 2019-528N; Ethics Committee II, Medical Faculty Mannheim, University of Heidelberg, Mannheim, Germany), according to the declaration of Helsinki. The HNSCC samples used in this study were collected from eight consenting patients with different tumour localisations (five in the oropharynx, two in the larynx and one in the hypopharynx), as listed in Table 1. Tissue procurement was carried out within 30 minutes of the surgical resection. The freshly resected sections were directly transferred to the Research and Development (R&D) Laboratory of the Department of Otorhinolaryngology, Head and Neck Surgery at University Hospital Mannheim (Mannheim, Germany) in a sterile standard tissue procurement medium. The procurement medium consisted of Dulbecco's Modified Eagle's Medium (DMEM; Bio Whittaker™ Classical Media BE 12-604FMB100, Lonza, Basel, Switzerland; from Biozym Scientific GmbH, Oldendorf, Germany, Cat. No. 880010) with 0.2% v/v puromycin 50 mg/ml (InvivoGen, Toulouse, France).

HNSCC ex vivo explant culture

The freshly procured tissues were washed twice with phosphate buffered saline (PBS; Gibco™ Cat. No. 14190144; Thermo Fisher Scientific, Inc.) and sliced into 6–9 pieces of approximately 3 mm thickness each. One portion of the tumour slice was immediately fixed in formalin and embedded in paraffin. This was considered the 'fresh' (i.e. uncultured) primary tissue control. The remaining tissue slices were cultured in a 12-well plate with inserts (ThinCert®; Greiner Bio-One,

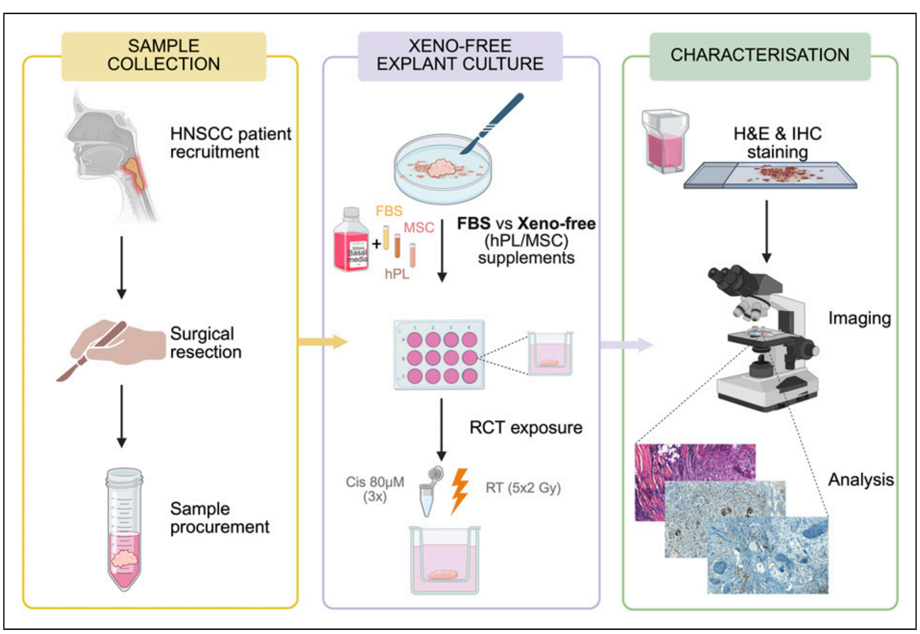

Figure 1. Schematic illustration of the steps involved in the establishment of the *ex vivo* Head and Neck Squamous Cell Carcinoma (HNSCC) explant model. FBS = fetal bovine serum; hPL = human platelet lysate; MSC = mesenchymal stem cell; Cis 80 μM = Cisplatin 80 μM; RT = radiotherapy; H&E = haematoxylin and eosin; IHC = immunohistochemical staining. [Created with BioRender.com.]

Table 1. Donor details of the tissues used to prepare HNSCC *ex vivo* cultures.

Donor	Age at initial diagnosis	Tumor localisation	TPS score (%)	Culture medium/ supplements used
A	69	Oropharynx	NC	FBS
B	59	Larynx	35	FBS
C	61	Oropharynx	5	FBS, hPL and MSC medium
D	62	Oropharynx	NC	FBS and hPL
E	70	Oropharynx	10	hPL and MSC medium
F	58	Oropharynx	20	hPL
G	76	Larynx	NC	MSC medium
H	63	Hypopharynx	NC	hPL

All tissues were from male donors. 'NC' denotes that the TPS score could not be calculated.

Frickenhausen, Germany), with the slices placed at the air–media interface (Figure 2). The slices were cultured in the various growth media under evaluation and were maintained in an incubator at 37°C with 5% v/v CO_2, as described in Affolter et al.[13]

Initially, the basal medium was prepared by adding 2% v/v L-glutamine (Lonza; Biozym Scientific GmbH, Cat. No. 882027-FFM) and 1% v/v penicillin/streptomycin (Gibco™ Cat. No. 15070063; Thermo Fisher Scientific, Inc.) to Lonza Bio Whittaker DMEM (as above). For the purposes of this study, the culture media for evaluation were prepared by adding one of the two supplements (i.e. FBS or hPL) to this basal medium, as detailed below:

— Fetal Bovine Serum (FBS)-supplemented medium was prepared by adding 10% v/v FBS (triple 0.1 μm filtered; Gibco™ Cat. No. A5256701; Thermo Fisher Scientific, Inc.) to the basal medium.
— Human platelet lysate (hPL)-supplemented medium was prepared by adding 12% v/v hPL and 0.02% v/v heparin (5000 U, Heparin Cofactor II, Human Plasma, Cat. No. 375115-100UG; Merck Millipore, France) to the basal media. The hPL was produced by pooling two buffy coat platelet concentrates; each platelet concentrate was produced by pooling four platelet-rich buffy coats from healthy blood donors (German Red Cross Blood Donor Service Baden-Württemberg–Hessen, Institute Mannheim, Germany). The platelet concentrates were lysed by a freezing and thawing step at −30°C and 37°C, followed by centrifugation at 2000*g* for 20 minutes at room temperature (RT). The supernatant was then aliquoted into sterile tubes and cryopreserved at −30°C. Prior to the experiments, the frozen

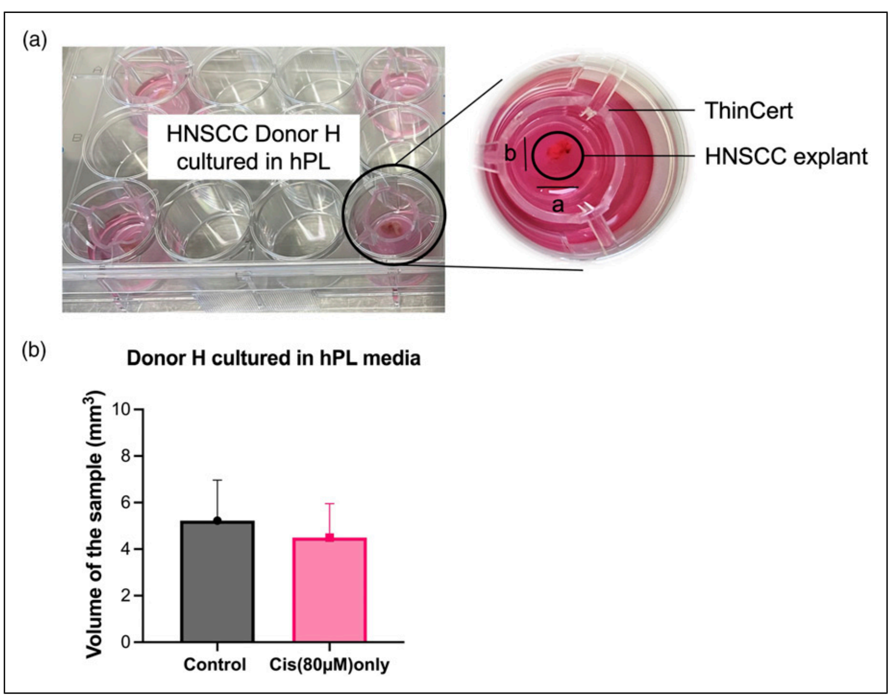

Figure 2. The experimental setup of the HNSCC explant culture model. (a) Schematic representation of the setup, showing that the volume of *ex vivo* samples remain unchanged following 80 μM Cis treatment on Day 3. (b) The sample size (expressed as mm^3 volume) was calculated by using the hemi-ellipsoid formula (Π/12 × diameter a × diameter b × Breslow tumour thickness), where horizontal-a, and vertical-b diameter measurements were taken. $n = 1$ (3 sections per donor, Donor H); Region of Interest (ROI) = 3 per section; hPL = human platelet lysate. Statistical analysis by using *t*-tests showed no significant effect of 80 μM Cis treatment on explant volume ($p = 0.2344$).

aliquots were thawed, centrifuged at 2000*g* for 10 minutes at RT and the required volume of supernatant was added to the basal medium to give a final concentration of 12%.[20,21]

— Commercially available Mesenchymal Stromal Cell (MSC) medium was used as a xeno-free control medium (StemMACS™ MSC Expansion Medium Kit XF, human; Miltenyi Biotec B.V. & Co. KG, Germany, Cat. No. 130-104-182) and prepared according to the manufacturer's protocol by adding 1.4% v/v StemMACS MSC Expansion Medium Supplement XF (human) to the StemMACS MSC Expansion Medium XF (human) basal medium. This particular commercial medium was selected for the study due to its well-characterised xeno-free formulation and its widespread use in MSC culture as a relevant comparator for assessing the efficacy of hPL-supplemented media in a clinically translatable context.

The different media used in this study, and their constituents, are described in Table 2. The HNSCC *ex vivo* tissue slices were grown in their respective growth medium; the media were changed every second day over the 10-day culture period.

Radiochemotherapy (RCT) treatment

The tissue slices were treated three times each, with the platinum-based chemotherapeutic drug cisplatin (S1166, Selleckchem, USA) at a concentration of 80 µM, on Days 1, 3 and 7 of the 10-day culture period.[13] Some of the cisplatin-treated tissue slices were also subjected to five irradiation doses on Days 1, 2, 3, 6 and 7. The RCT treatment was performed on three independent tissue slices (i.e. in triplicate). In parallel, certain slices were subjected to cisplatin treatment only or irradiation only as a comparison to investigate the effect of each treatment on the tissues. These treatments were also performed on three independent tissue slices (i.e. in triplicate).

The fractionated irradiation daily dose of 2 Gy was delivered with a medical linear accelerator with a photon acceleration potential of 6 MV (Synergy; Elekta AB, Stockholm, Sweden). The 12-well culture plate was positioned with 2 cm of polymethylmethacrylate slabs above and 5 cm below the plate, respectively. The fractionated radiation regimen was adapted according to the regimen applied to HNSCC patients in the clinic.[11]

Morphological evaluation of HNSCC ex vivo tissues

On Day 10, the tissue slices were harvested, fixed directly with 4% paraformaldehyde (Miltenyi Biotec B.V. & Co. KG; Cat. No. 1004960700) and paraffin-embedded in casketed blocks. After fixation, the formalin-fixed paraffin-embedded (FFPE) blocks were sectioned into 5-µm thick slices with a HISTOcut microtome (Leica AG, Germany). The sections were mounted onto standard adhesion glass slides that are commonly used for histological analysis (three sections/glass slide) and deparaffinised by washing three times in xylol for 5 minutes, followed by a serial ethanol wash (100%, 95%, 80%, 70%) for 5 minutes each. To visualise the morphology of the tissues, H&E staining was performed. All the reagents used for this purpose were purchased from Sigma-Aldrich (St Louis, MO, USA). The prepared glass slides were immersed in haematoxylin solution (HMM 999, Mayer's Haematoxylin, ScyTek Laboratories, USA) for 10–15 minutes and eosin (EYA 999, Eosin Y Solution, ScyTek Laboratories, USA) for 30 seconds. Then, the stained slides were mounted with mounting medium (HP68.1, ROTI® Mount, Carl Roth GmbH + Ko. KG, Germany) and coverslips, and imaged by using an optical microscope (Carl Zeiss AG, Germany).

Immunohistochemical staining

Immunohistochemical (IHC) staining was performed in order to characterise the cells in terms of their relative expression of the following biomarkers: Ki-67 (proliferation marker), PD-L1 (immune checkpoint marker) and vimentin (epithelial to mesenchymal transition (EMT) marker).

The slides were deparaffinised according to the procedure outlined in the previous section. Prior to staining, antigen retrieval ('demasking') was performed to unmask the epitopes that may have been masked during formalin fixation and paraffin embedding. This step enhances

Table 2. The growth media used and their constituents.

Growth medium	Base medium	Supplement	Antibiotics	Additional supplements
FBS-supplemented medium	DMEM + 2% v/v L-glutamine	10% v/v FBS	1% v/v penicillin/streptomycin	—
hPL-supplemented medium	DMEM + 2% v/v L-glutamine	12% v/v hPL	1% v/v penicillin/streptomycin	0.2% v/v heparin
MSC medium	StemMACS™ MSC expansion medium	Not applicable	Proprietary formulation	1.4% v/v StemMACS™ MSC expansion medium XF supplement

DMEM = Dulbecco's Modified Eagle's Medium; FBS = fetal bovine serum; hPL = human platelet lysate.

antibody binding during IHC. Accordingly, the deparaffinised slides were demasked with a buffer prepared in-house. Citrate-based (pH 6.0) or Tris-EDTA-based (pH 9.0) (Sigma-Aldrich) buffer was used, depending on the optimal retrieval conditions for the specific primary antibodies employed. The choice of buffer was based on the manufacturer's recommended protocol for each antibody to ensure maximum antigen accessibility. No additional components were added to the buffer used. Once the slides were demasked, they were incubated in 7% hydrogen peroxide (H_2O_2; Sigma-Aldrich) for 7 minutes and then blocked with 10% v/v normal sheep serum (Sigma-Aldrich) for 30 minutes. The slides were then incubated overnight at 4°C with the appropriate primary antibodies, diluted in commercially available antibody diluent (Cat. No. S080983-2; Dako, Agilent, Inc.). The primary antibodies were: anti-Ki-67 (1:200; Cat. No. IR626, Agilent Technologies, Inc.); anti-PD-L1 (1:200; Cat. No. 13684, Cell Signaling Technology, Inc.); anti-vimentin (1:200; Cat. No. 5741, Cell Signaling Technology, Inc.).

Following primary antibody incubation, the slides were washed three times in Dulbecco's balanced phosphate-buffered saline solution (DPBS — no calcium and no magnesium; ThermoFisher Scientific) for 5 minutes each wash. The samples were then incubated at 4°C for 30 minutes with secondary antibody: i.e. biotinylated anti-rabbit IgG antibody (1:200, Cat. No. RPN1004V; Cytiva, Marlborough, MA, USA) or anti-mouse IgG antibody (1:200, Cat. No. RPN1001V; Cytiva). After the secondary antibody incubation, the slides were again washed three times with DPBS. Subsequently, the samples were treated with biotinylated streptavidin horseradish peroxidase complex (Cat. No. RPN1051-2ML; Merck, Germany) for 45 minutes. For staining, 3-amino-9-ethylcarbazole (AEC) (ACG500-IFU, Scytek Laboratories, USA) was used. Precipitation of the substrate solution was monitored under the microscope and the sections were counterstained with haematoxylin (Sigma-Aldrich). The immuno-stained slides were mounted with mounting medium (SCY-PMT030, Permanent Mounting Media (Aqueous), Scytek Laboratories, USA) and a cover slip, and imaged with an optical microscope (Axiovert 25 CFL; Carl Zeiss AG, Germany). The images were processed and analysed by using Fiji software (ImageJ software).[22]

Tumour proportion score (TPS) and immunoreactive score (IRS)

The tumour proportion score (TPS), which is used for therapy decisions in clinical practice, was calculated for PD-L1 expression levels (Table 1). The following formula was used to determine the TPS:

$$\textit{Tumour proportion score}\ (TPS) = \frac{\textit{Number of positive membrane-bound stained tumour cells}}{\textit{Total number of vital tumour cells}}$$

For the evaluation of Ki-67 and vimentin expression levels, the immunoreactive score (IRS) was assessed. The IRS was determined from the IHC staining intensity scored by two independent observers,[13] and was an adaptation of Remmele and Stegner's method.[23]

$$\textit{Immunoreactive score}\ (IRS) = \textit{Staining intensity}\ (SI) \times \textit{Percentage positive cells}\ (PP)$$

Results

Comparison of FBS-supplemented growth medium with xeno-free alternatives

Initially, the *ex vivo* tissue culture conditions (i.e. the media volumes, supplement concentrations and frequency of media changes), as well as the RCT treatment regimen (i.e. the cisplatin concentration and radiation dose), were carefully optimised to achieve a system that preserved tissue viability and structure for at least 10 days, supported cellular proliferation, and maintained key biomarker expression representative of the tumour microenvironment. Ideally, the goal was to establish the system in all three culture media and to compare the tissue viability, proliferation and biomarker expression changes in response to RCT treatment. Due to the limited tissue availability, slices from all eight HNSCC primary tissues generally could not be cultured in the three types of media for comparison. The allocation of tissue slices to the different media was determined based on quantity and histological integrity of the available tissue following sectioning. Priority was given to ensuring that each experimental group had at least one representative donor for comparative analysis across all xeno-free culture media. However, due to better tissue availability, it was possible to compare the histological features of *ex vivo* tissue from Donor C following culture in all three media (see Table 1).

After 10 days of culture, a morphological evaluation was performed by H&E staining, in which the cultured tissues were compared to their corresponding native (i.e. untreated, uncultured and directly fixed) primary tissues, and after growth in the three different media. Due to better tissue availability, the tissue slices shown in Figure 3 for the three media comparisons are from Donor C. Tissue slices maintained in all three media were found to be similar in

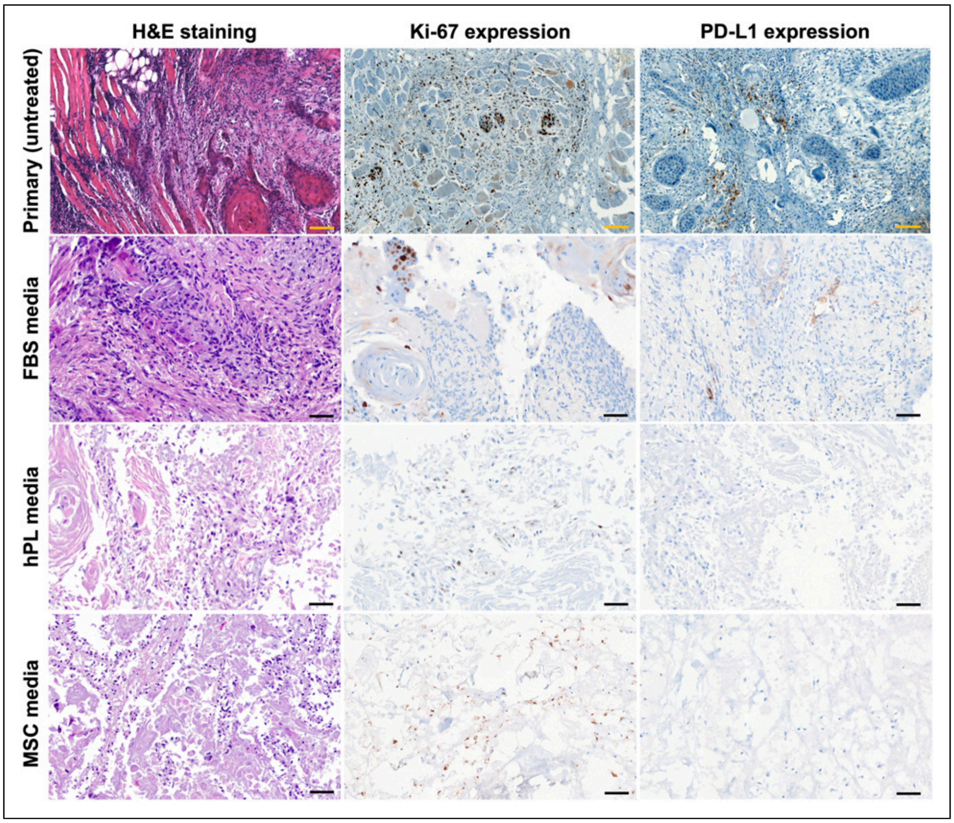

Figure 3. Morphological examination of HNSCC explants and primary tissues, cultured in the different media. The tissue explants were cultured in the respective medium for 10 days and their morphology was examined. The presence of Ki-67 and PD-L1 (both indicated by brown staining) was similar in the tissues cultured in all three media types and was consistently low. All tissues exhibited consistent representative tissue morphology (H&E staining), which was comparable between the tissues cultured in the different media and the untreated primary controls (Donor C); n = 3 technical replicates per condition. Scale bar = 100 and 50 μm for images from the primary samples and for the images from the different media, respectively.

histomorphology when compared to the primary tissue (Figure 3). In addition, the explants maintained in hPL-supplemented medium and in MSC medium displayed equivalent histomorphology to the tissues maintained in FBS-supplemented medium (Figure 3).

Having shown that, in principle, the HNSCC *ex vivo* tissues could be maintained in xeno-free medium, we proceeded with our analysis of the chosen biomarker expression levels via immunohistochemical staining. This analysis showed that tissue slices maintained in MSC medium exhibited relatively lower Ki-67 expression than those maintained in FBS-supplemented and hPL-supplemented media. PD-L1 expression was low in the untreated primary tissue, and this low level was similar in the tissues grown in all three media. In addition, the tissues maintained in MSC medium appeared morphologically distorted, as characterised by irregular architecture and disrupted epithelial integrity, and they also exhibited signs of necrosis (Figure 3). Since the primary tissue obtained from Donor C was taken from the oropharynx and had a TPS score of 5% (Table 1), this could explain the low PD-L1 and Ki-67 expression levels, and the morphologically distorted appearance of the tissue slices.

Effects of cisplatin on explants cultured in the different media

In a previous study based on the use of FBS-containing medium, we showed that the expression of PD-L1 is upregulated in response to treatment.[11] In the current study, we investigated whether the tissues maintained in xeno-free media would recapitulate these findings. Cisplatin treatment was performed on Days 1, 3 and 7, at the point of culture medium change. A treatment concentration of 80 μM cisplatin was used, following extensive dose range-finding experiments, and in line with our previous work.[11] The effects of cisplatin treatment on the *ex vivo* tissues maintained in the different media were assessed quantitatively (Figure 4). Specifically, we evaluated the comparative expression of PD-L1 (Figures 4(a) and (c)) and Ki-67 (Figures 4(b) and (d)), which is provided in terms of the tumour proportion score (TPS) for PD-L1 and percentage positive

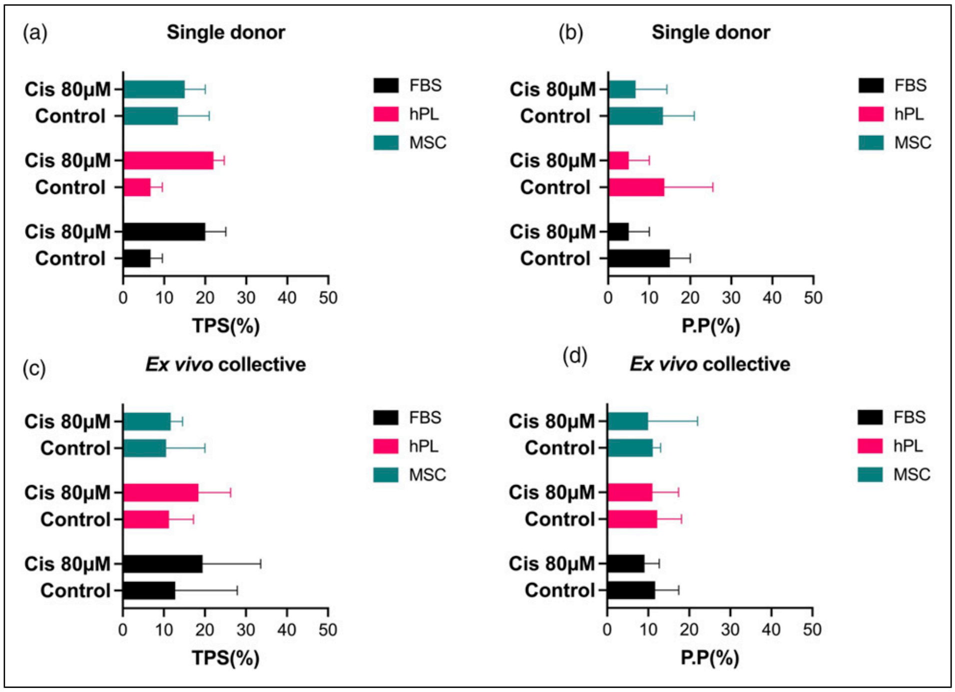

Figure 4. The effects of cisplatin treatment on PD-L1 and Ki-67 expression levels. The tumour proportion score (TPS), which reflects the relative presence of PD-L1 in the tissue, was assessed for tissues treated with cisplatin (and the untreated control) and cultured in the different media (FBS-supplemented medium, hPL-supplemented medium and MSC medium). PD-L1 data from a single donor are shown in (a), while data from the *ex vivo* collective samples are shown in (c). Cisplatin treatment versus the untreated control was compared across the three media; no significant differences were evident. The percentage positive cells (PP) score, which reflects the relative presence of Ki-67 in the tissue, was assessed for tissues treated with cisplatin (and untreated control) and cultured in the three different media. Ki-67 data from a single donor are shown in (b), while data from the *ex vivo* collective samples are shown in (d). Cisplatin treatment versus the untreated control was compared across the three media; no significant differences were evident. Single donor (a) and (b): n = 3 (3 sections per donor; Donor C was used). *Ex vivo* collective (c) and (d): n = 8 (3 sections per donor, Donors A to H were used). Region of Interest (ROI) = 3 per section. A two-way ANOVA analysis was performed.

cells (PP) score for Ki-67. The PP score is used for quantitative data interpretation rather than the immunoreactive score (IRS), because evaluation of staining intensity levels according to the Remmele and Stegner method[23] can be subjective.

Cisplatin treatment increased the relative expression of PD-L1, and this trend was consistent across all three media. However, the magnitude of PD-L1 upregulation was less pronounced in tissue grown in MSC medium, possibly due to donor-specific tissue variability, reduced drug sensitivity, or altered stress response in this medium. The type of medium used did not have a significant effect after exposure of the tissues to cisplatin (18.35% total variance, significant p value of 0.0246; comparison between cisplatin-treated and untreated control tissues across all media types). There was an evident increase in PD-L1 expression in cisplatin-treated tissues, as compared to the untreated control tissues (p value = 0.0007, with a total variance of 51.64%) (Figure 4(a)).

Expression of Ki-67 (proliferation biomarker) was similar in all the cultured samples, across all three media. However, there was a slight reduction in proliferation when the tissues were treated with cisplatin, as compared to the untreated control tissues. The percentage of Ki-67-positive cells significantly decreased following cisplatin exposure (32.25% total variance, p value = 0.0045; comparison between cisplatin-treated and untreated control tissues across all media types (Figure 4(b)). The type of medium did not significantly affect the Ki-67 expression levels; the decrease in proliferation following cisplatin exposure was consistently observed in tissues cultured in all media types.

In parallel, we observed a similar pattern in the TPS and PP values discerned from all the *ex vivo* tissue collective groups treated with 80 μM cisplatin (Figures 4(c) and (d)). Although differences were subtle, the quantitative data showed a consistent, if modest, decrease in Ki-67 following cisplatin treatment, even if this was not as visually striking in some media, such as MSC medium. Overall, the type of medium used to culture the HNSCC tissues did not have any significant effects on PD-L1 expression or on the proliferation rate, as determined by Ki-67 expression. Hence, it can be concluded that both types of xeno-free media (i.e. the hPL-supplemented medium and the MSC medium) could potentially be used as alternatives to FBS-supplemented medium for the culture of HNSCC tissues.

We also assessed tissue morphology after cisplatin treatment (Figure 5). The explants maintained in FBS-supplemented medium and in the xeno-free media (i.e. hPL-supplemented medium and MSC medium) underwent similar morphological changes when exposed to cisplatin. In the untreated controls, the cells appeared cuboidal, which is a characteristic feature of squamous cells (Figure 5).[24–26] The tissues treated with cisplatin appeared to be circular and in distress (Figure 5).[2,25] This observation correlated with the distorted morphology, loss of epithelial architecture and early signs of necrosis observed in some cisplatin-treated tissues — particularly those cultured in MSC medium, as reported in Figure 3.

Effects of radiochemotherapy (RCT) treatment on HNSCC explants cultured in xeno-free media

We previously investigated the effects of irradiation on the expression of PD-L1 in HNSCC *ex vivo* tissues, as compared to 2-D *in vitro* HNSCC cultures.[11] Here, as a proof-of-principle, we exposed the *ex vivo* tissues to fractionated irradiation of 2 Gy, on five days (i.e. Day 1, 2, 3, 6 and 7). In addition to radiation or cisplatin treatment only, some tissue samples were exposed first to 80 μM cisplatin followed by radiation. Due to the limited amount of available tissues, the simulation of RCT treatment was performed only with tissues cultured in the xeno-free media (i.e. hPL-supplemented medium and MSC medium). As expected, after exposure to radiation the tissues underwent morphological changes (Figure 6(a)), and enlarged cells were observed in the irradiated tissues. Tissue explants that had been also treated with cisplatin appeared necrotic/apoptotic. The observed changes in cell morphology were similar in tissues cultured in both hPL-supplemented medium and MSC medium.

Furthermore, after RCT treatment there was an upregulation in PD-L1 expression, as well as a decrease in Ki-67-positive cells. This result was evident in the immunohistochemistry-stained images (Figures 6(b) and (c)). The tissues in both hPL-supplemented medium and MSC medium exhibited similar features in this respect. Quantification of the results revealed that there was no significant difference in PD-L1 expression after RCT treatment between tissues grown in hPL-supplemented medium or MSC medium (Figure 7(a)). Although there was no significant difference in PD-L1 expression between irradiated tissues grown in hPL-supplemented medium or in MSC medium, we observed a significant increase in PD-L1 expression in tissues grown in hPL-supplemented medium when exposed to 80 μM cisplatin only, as compared to the untreated control tissues (see section above). The increase was visually evident; however, statistically, it was not considered to be an effective upregulation. This insignificant change was observed in specimens cultured in both hPL-supplemented medium and MSC medium. Overall, while treatment variation accounted for 34.42% of the total variance in PD-L1 expression ($p = 0.0001$), the type of medium had a greater impact, contributing to 54.52% of the total variance ($p = 0.0011$).

When Ki-67 expression was assessed (via the PP score), it showed a significant drop when the tissues had been treated with cisplatin and radiation (Cis (80 μM) + RT) (Figure 7(b)). This extremely significant decrease was consistent in tissues grown in both hPL-supplemented medium and MSC medium. Although cisplatin treatment alone did not result in a significant decrease in Ki-67

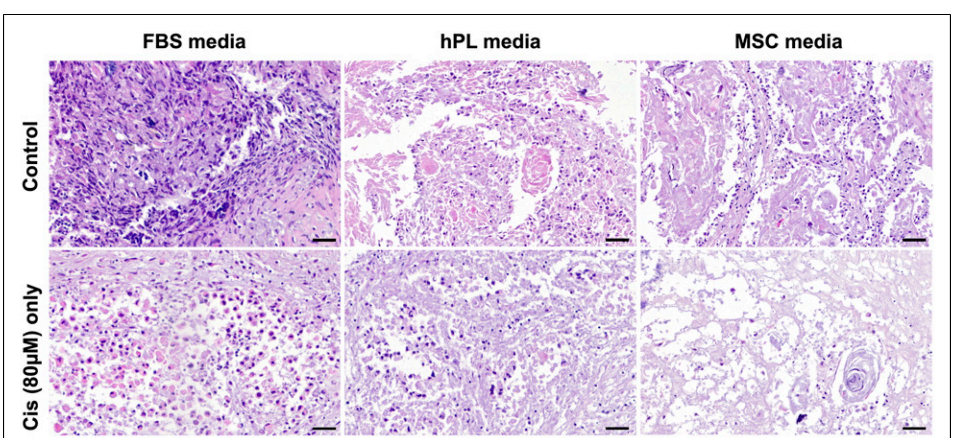

Figure 5. Morphological examination of HNSCC explants cultured in different media and treated with cisplatin. The tissues, cultured in either FBS-supplemented medium, hPL-supplemented medium or MSC medium, were treated with 80 μM cisplatin (Cis). General morphology can be examined after H&E staining. The typical characteristic HNSCC cell shape changed into a rounded shape after cisplatin treatment. This was consistently observed for all tissues. Scale bar = 50 μm for all images.

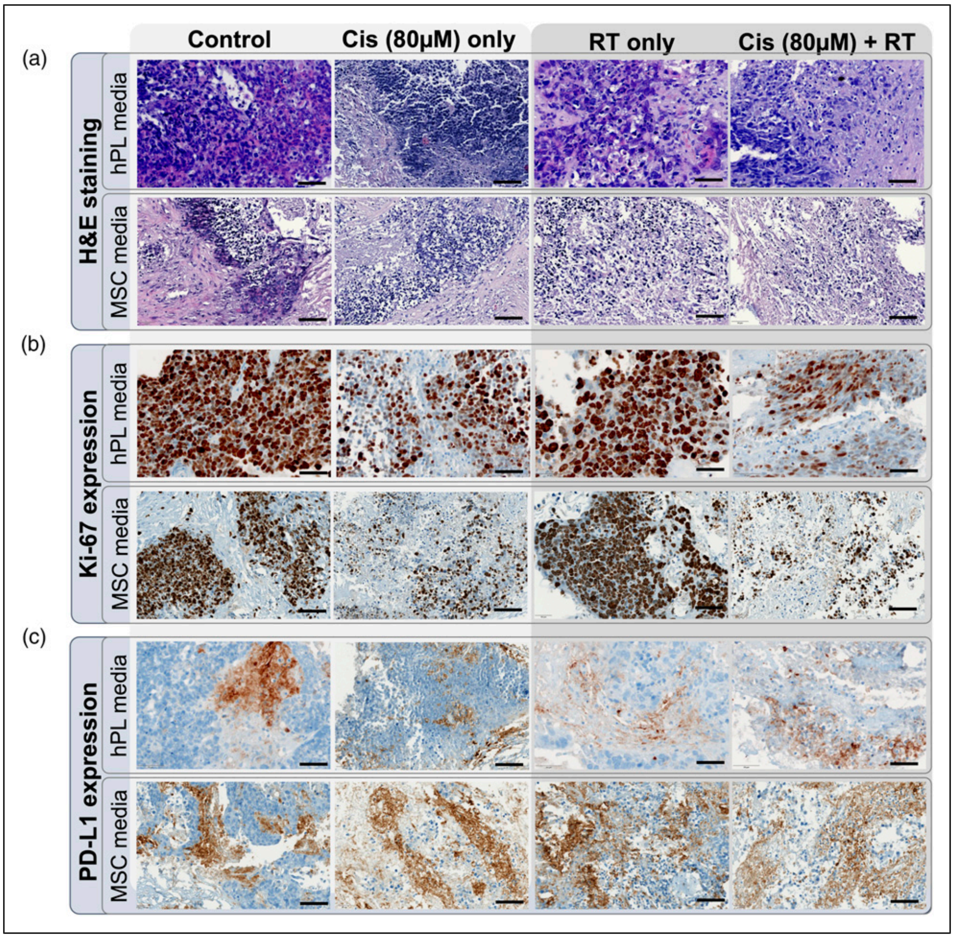

Figure 6. Haematoxylin and eosin (H&E) and immunohistochemical (IHC) staining of xeno-free cultured tissues treated with chemotherapy and radiotherapy. The tissues, cultured in either hPL-supplemented medium or MSC medium, were treated with 80 μM cisplatin (Cis), radiotherapy (RT) or a combined regimen of Cis plus RT. General morphology can be seen in (a) after H&E staining. Immunohistochemical analysis shows the presence of (b) Ki-67 and (c) PD-L1. $n = 2$ (3 sections per donor, Donors F and G); Region of Interest (ROI) = 3 per section; scale bar = 50 μm for all images.

expression for the tissues grown in the MSC medium, radiation therapy (RT) only and RCT led to significant decreases in Ki-67 expression. Additionally, while the type of medium had a notable influence on PD-L1 expression, it accounted for only 8.78% of the total variance in Ki-67 expression ($p = 0.0083$). In contrast, for Ki-67 levels, the type of treatment was the predominant factor, contributing to 80.16% of the total variance ($p < 0.0001$).

Finally, the *ex vivo* tissues in xeno-free media were characterised to determine whether radiation treatment influences the tissues to undergo EMT transition, as determined by their vimentin expression levels (Figure 8). In comparison to the untreated control tissues, decreased vimentin expression was evident after RCT treatment, and this was similarly observed in tissues maintained in the hPL-supplemented medium and in the MSC medium. Tissues cultured in MSC medium that were exposed to cisplatin only, and those subjected to cisplatin plus radiation treatment (RCT) appeared to be necrotic and to have disintegrated.

Discussion

The aim of this study was to determine whether, in principle, FBS-supplemented medium could be replaced with xeno-free alternatives for use in HNSCC explant culture. In practice, both options assessed in this study, hPL-supplemented medium and MSC medium, have been tested as alternatives to the widely-used FBS-supplemented medium.[21,27,28] Lensch et al.[29] reported to have improved the yield of viable mesenchymal cells with the use of hPL-supplemented medium and MSC medium in 2-D cultures. However, with respect to the goal of the current study, the challenge was to maintain cellular heterogeneity of the HNSCC tissues in longer-term culture, so as to best reflect the complexity and architecture of the original native primary tissue. In addition, xeno-free culture models must be

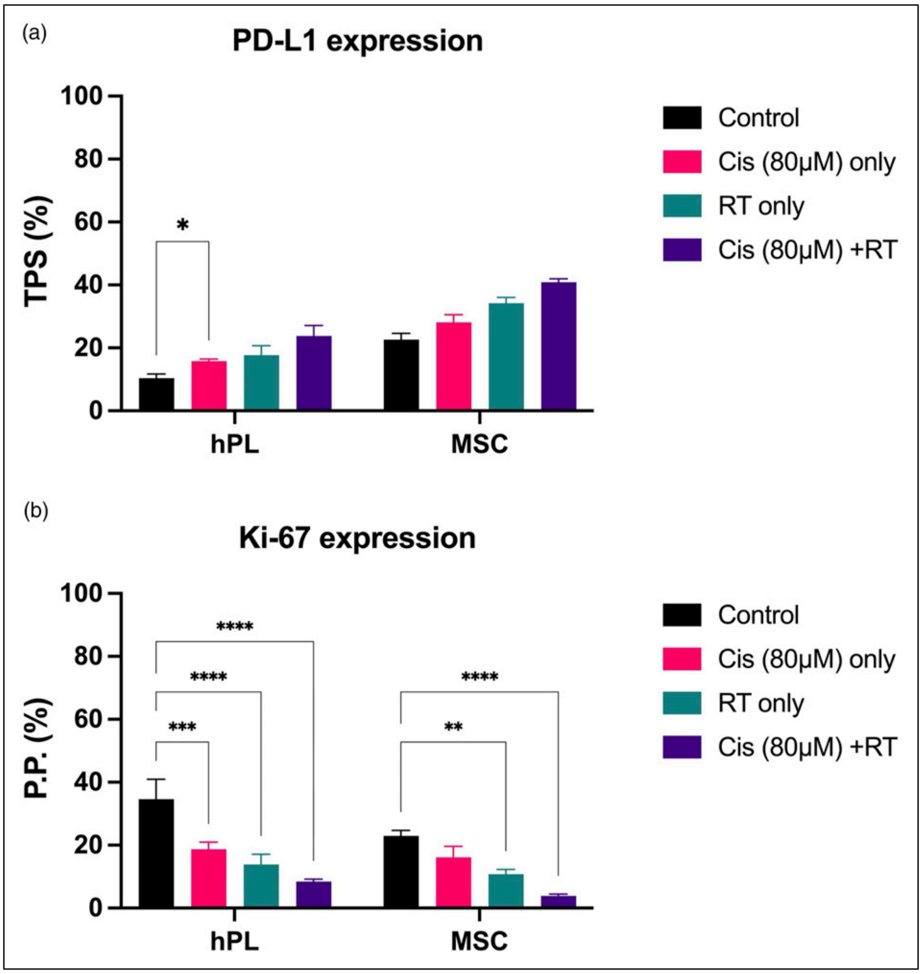

Figure 7. The effects of chemotherapy and radiotherapy treatment on PD-L1 and Ki-67 levels. The tissues, cultured in either hPL-supplemented medium or MSC medium, were treated with 80 μM cisplatin (Cis), radiotherapy (RT) or a combined regimen of Cis plus RT. Immunohistochemistry was performed to assess (a) the tumour proportion score (TPS), which reflects the relative presence of PD-L1 in the tissue; and (b) the percentage positive cells (PP) score, which reflects the relative presence of Ki-67 in the tissue. A two-way ANOVA was performed, $*p < 0.05$, $**p < 0.005$, $***p < 0.0005$, $****p < 0.0001$; $n = 2$ (3 sections per donor, Donors F and G); Region of Interest (ROI) = 3 per section.

versatile, in order to permit the efficacy testing of specific treatment regimens, e.g. personalised medicine approaches. As an initial step, we worked with our already established HNSCC *ex vivo* model, which can be successfully cultured for 10 days according to our previous data. Figure 2 shows the experimental set-up and highlights the size (volume) of the tissue cultured within the well.[11,13]

In the current study, the initial histomorphology results depicted comparable morphological traits in explants maintained in the xeno-free media, explants maintained in FBS-containing medium, and primary (uncultured) tissue. In addition, Ki-67 and PD-L1 expression levels were consistent across all three media and were similar to those in the respective primary tissues. Furthermore, it was revealed that the HNSCC explants could be grown viably *ex vivo*, retaining their proliferation capacity for up to 10 days in the xeno-free hPL-supplemented medium and MSC medium.

As observed in our previous study on *ex vivo* 2-D and 3-D cultures grown in FBS-containing media,[11] the explants grown in xeno-free media in the current study exhibited morphological changes, as well as upregulation of PD-L1 and a decrease in proliferation (Ki-67 expression) in response to irradiation. To further advance our explant model to ensure that it can best reflect clinical conditions, and to permit comparison with existing 2-D model data, we treated the explants with RCT. Given that the tissues in both xeno-free media demonstrated behaviour similar to those grown in FBS-containing medium, and considering the limited availability of tissue samples, we proceeded to culture the HNSCC tissues solely in xeno-free media and expose them to RCT.

We observed upregulation of PD-L1 expression in tissues cultured in both xeno-free media following RCT treatment. These findings were consistent across tissues grown in both

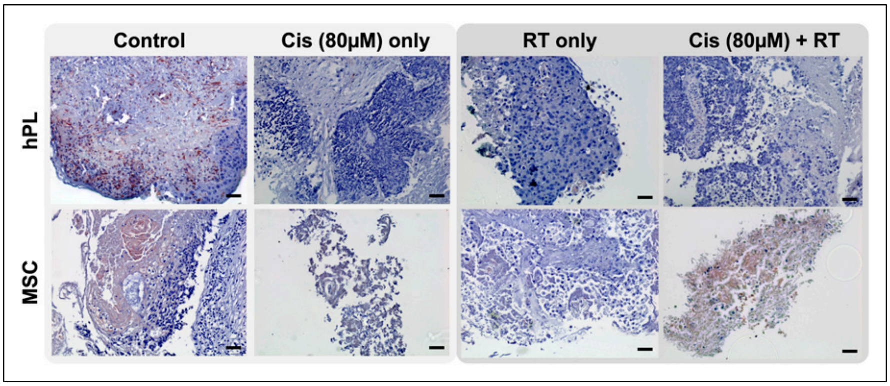

Figure 8. The effects of chemotherapy and radiotherapy treatment on vimentin levels. The tissues, cultured in either hPL or MSC medium, were treated with 80 μM cisplatin (Cis), radiotherapy (RT) or a combined regimen of Cis plus RT. Immunohistochemistry was performed to assess the levels of vimentin, a marker of epithelial to mesenchymal transition. *n* = 3; Region of Interest (ROI) = 3 per section; scale bar = 50 μm for all images.

hPL-supplemented medium and MSC medium. However, the extent to which PD-L1 was expressed at the basal level was higher in MSC medium than in hPL-supplemented medium. To date, studies on the impact of PD-L1 expression in response to RCT treatment have reported varying findings. Significant upregulation of PD-L1 was observed in certain studies, while others reported significant down-regulation, particularly in response to platinum-based therapy combined with irradiation.[11,30,31]

A previous study[11] conducted by our research group on *ex vivo* 2-D and 3-D culture models found that there was a significant, dose-dependent increase in PD-L1 expression levels in irradiated cells, with this effect being further amplified following RCT treatment. Our findings in the current study on the HNSCC explant cultures are consistent with and comparable to these previous observations. However, PD-L1 expression exhibited variability between tissue slices maintained in MSC and hPL-supplemented media, which may be attributed to the distinct composition of growth factors and cytokines in each medium type. As the changes in expression levels in response to RCT were comparable between the two cell culture media groups, we assume that the proportional increase in the PD-L1 expression was more prominent in hPL-supplemented medium than in the MSC medium-cultured group. This could be due to a lower baseline PD-L1 expression in the hPL group prior to treatment, leading to a prominent relative increase. Given that hPL is derived from human platelets, it contains bioactive molecules that could influence immune-related signalling pathways, potentially leading to differential PD-L1 regulation.[18] Accordingly, treatment variation is higher due to an induction of PD-L1 by the MSC medium. This induction has already been demonstrated in breast cancer cells, and is primarily mediated by the secretion of cytokines (such as CCL5).[32] This highlights the critical role of the tumour microenvironment in modulating immune responses and underscores the importance of selecting appropriate culture conditions. To substantiate the initial observations in this preliminary study, a significantly larger cohort must be selected from patients with HNSCC. However, this would need to be carried out in a future study, as in the current study the sample amounts were limiting, and our main aim was the establishment of a xeno-free HNSCC explant model.

Concurrently, a significant reduction in proliferation rate was observed in response to both irradiation and RCT treatment for tissues cultured in both xeno-free media. Notably, the extent of Ki-67 downregulation was greater in hPL-supplemented medium compared to those in MSC medium-cultured tissues. However, MSC medium-cultured samples exhibited low Ki-67 expression (~5%) following RCT compared to those cultured in hPL-supplemented medium. In contrast, hPL-cultured tissues indicated a more robust proliferative decline to both irradiation and RCT. A particularly significant decrease in proliferation level was observed in response to RCT treatment, with irradiation demonstrating a more pronounced inhibitory effect on proliferation than cisplatin chemotherapy alone. This trend was consistent across tissues cultured in both hPL-supplemented medium and MSC medium. It is possible that RCT treatment might trigger apoptosis, as evidenced by the significant reduction in proliferation.[13,33,34] To determine the mechanism of cell death and elucidate whether the observed decrease in proliferation is due to apoptosis or necrosis, additional investigations focusing on epithelial growth factor receptor (EGFR) biomarkers could be performed.[30,34] Although treatment variations significantly impacted tissue viability and proliferation, the same patterns were observed for tissue slices cultures in both types of xeno-free media.

This study also assessed the presence of vimentin, a marker of epithelial-to-mesenchymal transition (EMT). The radiation-induced modulation of vimentin expression in HNSCC remains poorly understood, and has not been thoroughly investigated in the literature to date.[33,35] HNSCC undergoing partial-EMT is considered a prognostic marker; this is a characteristic of tumour cell migration and metastasis, and is associated with poor-prognosis. HNSCCs have generally been reported to undergo partial-EMT, which appears to be site-dependent, occurring in distinct cellular populations within specific tumour regions.[36] Our data are in contrast to this theory, as we observed a decrease in vimentin in all samples and across all media, which could indicate the effectiveness of the RCT treatment. However, this requires further analysis, possibly involving a larger number of samples from a diverse patient cohort. Future studies could include single-cell transcriptome analysis and the investigation of E-cadherin (a characteristic epithelial biomarker) expression dynamics throughout RCT treatment in combination with vimentin expression dynamics analysis.

Several studies have explored the potential of xeno-free alternatives to FBS, with particular focus on hPL and commercially available MSC-specific media. These studies compared naturally-derived hPL to commercially available MSC media and supplements, with a particular focus on their effects on MSC expansion and functionality. For instance, an investigation into GMP-compliant protocols demonstrated that hPL can effectively replace FBS for MSC expansion, achieving higher cell yields without compromising key MSC characteristics, supporting the suitability of hPL for clinical applications.[37] Similarly, another study compared clinical grade hPL with FBS for culturing MSCs from bone marrow and adipose tissue. hPL accelerated MSC proliferation without causing chromosomal abnormalities, and the cells maintained their differentiation potential, highlighting hPL as a viable alternative to FBS.[38] While these findings support our findings in highlighting the feasibility of using xeno-free conditions for HNSCC explants, further work is needed to refine the model and address practical challenges associated with its broader application.

We do acknowledge the importance of extending this investigation to a significantly larger patient cohort, conducting comprehensive histochemical analyses to evaluate PD-L1 expression modulation during RCT treatment, as well as parallel assessment of vimentin and E-cadherin expression dynamics. Additionally, it is essential to determine whether the *ex vivo* culture model accurately reflects clinical outcomes. Given that our primary objective is to refine the model (by optimising physiological relevance) and replace xeno-derived supplements, our future efforts will focus on transitioning to fully animal-free/xeno-free antibodies and using xeno-free supplements throughout the entire experimental process. Implementing this approach will require a rigorous, step-by-step optimisation process, with careful comparison to established methods through the use of strategies such as H&E and IHC staining. In addition, given that HNSCCs are localised in the oral cavity, an area highly exposed to environmental contaminants, only 30% of the *ex vivo* samples collected from patients survived culture and experimentation without any microbial contamination. This situation further exacerbates the already existing challenge, i.e. the limited availability of tumour tissue samples, particularly for HNSCCs. To address this issue, establishing a cohort study across different regions and clinics would help to overcome the limitation of tissue availability. Moreover, it would provide a foundation for other HNSCC research units to develop xeno-free models.

Conclusions

In summary, we have successfully established an HNSCC tissue explant model based on the use of xeno-free media, which has the potential to replace the widely used FBS-supplemented media. Importantly, we were able to mimic standard treatment of HNSCC with our *ex vivo* model — namely, chemotherapy combined with radiotherapy, which is essential for studying physiologically relevant human-specific cellular mechanisms of treatment sensitivity and resistance. This approach aligns with the principles of ethical research and the Three Rs (*replacement*, *reduction* and *refinement*), providing a viable alternative to the use of animal models that upholds scientific rigour while minimising the use of xeno-derived components.

Ultimately, this comprehensive investigation will not only enhance the refinement of our HNSCC explant model but also pave the way for the development of a truly non-human animal-free system. Such a human-focused system would be of great relevance for precision medicine, particularly for the development of innovative and novel therapeutic strategies that require a human-specific testing platform. Moreover, the HNSCC explant model established in xeno-free media alternatives could serve as a unique tool for drug sensitivity testing, serving as a unique platform to validate novel treatments and explore treatment options. By bridging the gap between preclinical studies and clinical applications, such a xeno-free tumour model would contribute significantly to advancing precision oncology, ultimately benefiting patient care through more tailored and effective therapeutic interventions.

Acknowledgements

We extend our gratitude to the surgeons of the Department of Otorhinolaryngology, Head and Neck Surgery, Mannheim University Hospital, for their contribution in segregating resected tumour samples for research and pathological evaluation. We would like to thank Prof. Karen Bieback and her laboratory (Institute of Transfusion Medicine and Immunology, Medical Faculty Mannheim, Germany) for sharing the human platelet lysate (hPL) and the hPL extraction protocol for our research purposes. We also acknowledge the Institute of Pathology, Mannheim University Hospital, for their work in histological staining and imaging. Furthermore, we appreciate the technical support provided by Ms

Petra Prohaska, Department of Otorhinolaryngology, Head and Neck Surgery, Medical Faculty Mannheim, Heidelberg University.

ORCID iDs

Alexya Azhakesan https://orcid.org/0000-0003-2482-5974
Annette Affolter https://orcid.org/0000-0002-2604-3020

Authors' contributions

AAz and ES were in charge for the experimentation, data acquisition and formal analysis under the assistance of LH and esteemed guidance of AA, JK and NR. AA conceived the study. ES, SL, FJ, LB, CS, AL and AA were responsible for curating and coordinating the consented patient cohort. ES and AAz determined their treatment regimen. CB contributed to pathological analyses. JF enabled the irradiation experiments. AA and NR were responsible for funding acquisition. AAz, ES, AA, LH and JK were responsible for the figures. AAz, ES and AA wrote the original draft of the manuscript. AAz, ES, AA, JK and NR were responsible for writing, reviewing and editing the manuscript. NR and AA confirm the authenticity of all the raw data. All authors read and approved the final manuscript.

Funding

The research on 3-D models of head and neck squamous cell carcinoma (HNSCC) has been funded by the Ministry of Rural Affairs, Food, and Consumer Protection of Baden-Wuerttemberg, Germany, under the research-funding programme 'Development of Replacement and Complementary Methods to Reduce Animal Testing'. This funding was awarded to Prof. Dr Annette Affolter (Staatshaushaltsplan 2020/2021 Kap. 0802, Tit. Gr. 74). Additional support was provided by the 3R network Baden-Wuerttemberg, funded by the Ministry of Science, Baden-Wuerttemberg, Germany, to Prof. Dr Nicole Rotter and Prof. Dr Karen Bieback. The 3R network grant (33-7533-6-1522/10/4) also supports the contributions of Alexya Azhakesan.

Declaration of conflicting interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data availability statement

All data generated or analysed during this study are included in this publication.

References

1. Arutyunyan I, Jumaniyazova E, Makarov A, et al. *In vitro* models of head and neck cancer: From primitive to most advanced. *J Pers Med* 2023; 13: 1575.
2. Wang G, Zhang M, Cheng M, et al. Tumor microenvironment in head and neck squamous cell carcinoma: Functions and regulatory mechanisms. *Cancer Lett* 2021; 507: 55–69.
3. Glastonbury CM, Parker EE and Hoang JK. The postradiation neck: Evaluating response to treatment and recognizing complications. *AJR Am J Roentgenol* 2010; 195: W164–W171.
4. Curry JM, Sprandio J, Cognetti D, et al. Tumor microenvironment in head and neck squamous cell carcinoma. *Semin Oncol* 2014; 41: 217–234.
5. Chaves P, Garrido M, Oliver J, et al. Preclinical models in head and neck squamous cell carcinoma. *British J Cancer* 2023; 128: 1819–1827.
6. Canning M, Guo G, Yu M, et al. Heterogeneity of the head and neck squamous cell carcinoma immune landscape and its impact on immunotherapy. *Front Cell Dev Biol* 2019; 7: 52.
7. Bernal PN. *Crafting tissue complexity: Pioneering layerwise and volumetric biofabrication strategies for advanced in vitro models*. UMC Repository: Utrecht University, 2024.
8. Bonartsev AP, Lei B, Kholina MS, et al. Models of head and neck squamous cell carcinoma using bioengineering approaches. *Crit Rev Oncology Hematol* 2022; 175: 103724.
9. Méry B, Rancoule C, Guy J-B, et al. Preclinical models in HNSCC: A comprehensive review. *Oral Oncol* 2017; 65: 51–56.
10. Lin CJ, Grandis JR, Carey TE, et al. Head and neck squamous cell carcinoma cell lines: Established models and rationale for selection. *Head Neck* 2007; 29: 163–188.
11. Affolter A, Liebel K, Tengler L, et al. Modulation of PD-L1 expression by standard therapy in head and neck cancer cell lines and exosomes. *Int J Oncol* 2023; 63: 102.
12. Engelmann L, Thierauf J, Koerich Laureano N, et al. Organotypic co-cultures as a novel 3D model for head and neck squamous cell carcinoma. *Cancers* 2020; 12: 2330.
13. Affolter A, Muller M-F, Sommer K, et al. Targeting irradiation-induced mitogen-activated protein kinase activation *in vitro* and in an *ex vivo* model for human head and neck cancer. *Head Neck* 2016; 38(S1): E2049–E2061.
14. Yao T and Asayama Y. Animal-cell culture media: History, characteristics, and current issues. *Reprod Med Biol* 2017; 16: 99–117.
15. Liu S, Yang W, Li Y, et al. Fetal bovine serum, an important factor affecting the reproducibility of cell experiments. *Sci Rep* 2023; 13: 1942.
16. Grimm H, Biller-Andorno N, Buch T, et al. Advancing the 3Rs: Innovation, implementation, ethics and society. *Front Vet Sci* 2023; 10: 1185706.
17. Cederroth CR and Sandström J. Editorial: Towards a new 3Rs era in experimental research. *Front Behav Neurosci* 2024; 18: 1404294.
18. Rauch C, Feifel E, Amann E-M, et al. Alternatives to the use of fetal bovine serum: Human platelet lysates as a serum substitute in cell culture media. *ALTEX* 2011; 28: 305–316.
19. Duarte AC, Costa EC, Filipe HAL, et al. Animal-derived products in science and current alternatives. *Biomater Adv* 2023; 151: 213428.
20. Bieback K. Platelet lysate as replacement for fetal bovine serum in mesenchymal stromal cell cultures. *Transfus Med Hemother* 2013; 40: 326–335.
21. Bieback K, Fernandez-Muñoz B, Pati S, et al. Gaps in the knowledge of human platelet lysate as a cell culture supplement for cell therapy: A joint publication from the AABB

and the International Society for Cell & Gene Therapy. *Cytotherapy* 2019; 21: 911–924.
22. Schindelin J, Arganda-Carreras I, Frise E, et al. Fiji: An open-source platform for biological-image analysis. *Nat Methods* 2012; 9: 676–682.
23. Remmele W and Stegner HE. Recommendation for uniform definition of an immunoreactive score (IRS) for immunohistochemical estrogen receptor detection (ER-ICA) in breast cancer tissue. *Pathologe* 1987; 8: 138–140.
24. Fernandez-Rebollo E, Mentrup B, Ebert R, et al. Human platelet lysate versus fetal calf serum: These supplements do not select for different mesenchymal stromal cells. *Sci Rep* 2017; 7: 5132.
25. Panella S, Muoio F, Jossen V, et al. Chemically defined xeno- and serum-free cell culture medium to grow human adipose stem cells. *Cells* 2021; 10: 466.
26. González Hernández Y and Fischer RW. Serum-free culturing of mammalian cells — adaptation to and cryopreservation in fully defined media. *ALTEX* 2007; 24: 110–116.
27. Hemeda H, Giebel B and Wagner W. Evaluation of human platelet lysate versus fetal bovine serum for culture of mesenchymal stromal cells. *Cytotherapy* 2014; 16: 170–180.
28. Guiotto M, Raffoul W, Hart AM, et al. Human platelet lysate to substitute fetal bovine serum in hMSC expansion for translational applications: A systematic review. *J Transl Med* 2020; 18: 351.
29. Lensch M, Muise A, White L, et al. Comparison of synthetic media designed for expansion of adipose-derived mesenchymal stromal cells. *Biomedicines* 2018; 6: 54.
30. Steen S, Semmelmayer K, Flechtenmacher C, et al. Dynamic up-regulation of PD-L1 in the progression of oral squamous cell carcinoma. *Int J Mol Sci* 2023; 24: 16386.
31. Akinleye A and Rasool Z. Immune checkpoint inhibitors of PD-L1 as cancer therapeutics. *J Hematol Oncol* 2019; 12: 92.
32. Aboulkheyr Es H, Bigdeli B, Zhand S, et al. Mesenchymal stem cells induce PD-L1 expression through the secretion of CCL5 in breast cancer cells. *J Cell Physiol* 2021; 236: 3918–3928.
33. Zamulaeva I, Matchuk O, Selivanova E, et al. Effects of fractionated radiation exposure on vimentin expression in cervical cancers: Analysis of association with cancer stem cell response and short-term prognosis. *Int J Mol Sci* 2023; 24: 3271.
34. Rong C, Muller MF, Xiang F, et al. Adaptive ERK signalling activation in response to therapy and *in silico* prognostic evaluation of EGFR-MAPK in HNSCC. *Br J Cancer* 2020; 123: 288–297.
35. Liu S, Liu L, Ye W, et al. High vimentin expression associated with lymph node metastasis and predicated a poor prognosis in oral squamous cell carcinoma. *Sci Rep* 2016; 6: 38834.
36. Pal A, Barrett TF, Paolini R, et al. Partial EMT in head and neck cancer biology: A spectrum instead of a switch. *Oncogene* 2021; 40: 5049–5065.
37. Becherucci V, Piccini L, Casamassima S, et al. Human platelet lysate in mesenchymal stromal cell expansion according to a GMP grade protocol: A cell factory experience. *Stem Cell Res Ther* 2018; 9: 124.
38. Juhl M, Tratwal J, Follin B, et al. Comparison of clinical grade human platelet lysates for cultivation of mesenchymal stromal cells from bone marrow and adipose tissue. *Scand J Clin Lab Invest* 2016; 76: 93–104.